

Evidence before leaderboard claims

A repaired evidence contract turns a contaminated benchmark and a failed hypothesis into defensible research.

MiT-B3 U-Net versus nnU-Net v2

Negative contour-channel ablation
Protected test remains sealed; this is not state of the art

EVIDENCE
ATLAS

Leakage changed the meaning of every favorable score

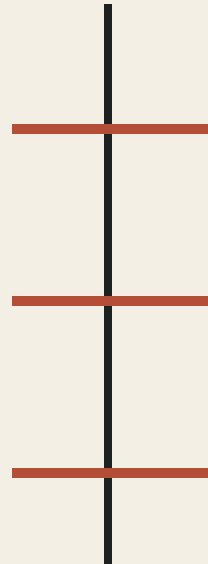
Three patient identities and 13 rows crossed legacy split boundaries, so the old leaderboard became historical evidence rather than a scientific ranking.

3

crossing patient IDs

13

cross-boundary rows



- Patient overlap violates independent patient-level interpretation.
- Old test-v2 values remain visible only as contamination-aware history.
- The repair starts with identity groups, not model tuning.

Two hypotheses, two evidence classes

The architecture comparison and the contour-channel ablation answer different questions and must not be collapsed into one ranking.

RQ1

How do complete MiT-B3 and nnU-Net systems trade overlap, boundary quality, precision, and recall?

System comparison
Single recorded run
Descriptive inference

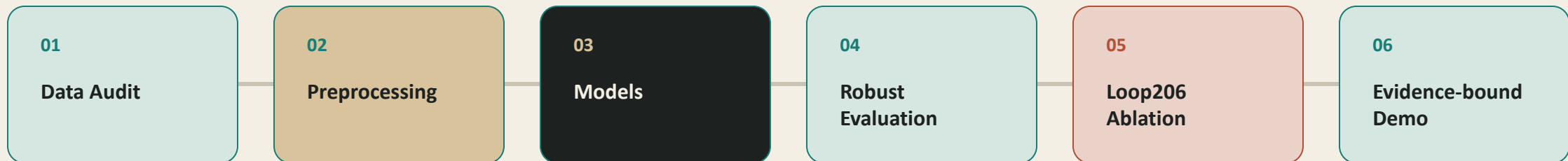
RQ2

Does a saliency-constrained contour channel improve a matched MiT-B3 control?

Matched intervention
Three selected seeds
Conditional paired inference

The claim can only be as strong as its weakest evidence boundary

Every model output passes through data, protocol, metric, and claim gates before it reaches the paper or demo.



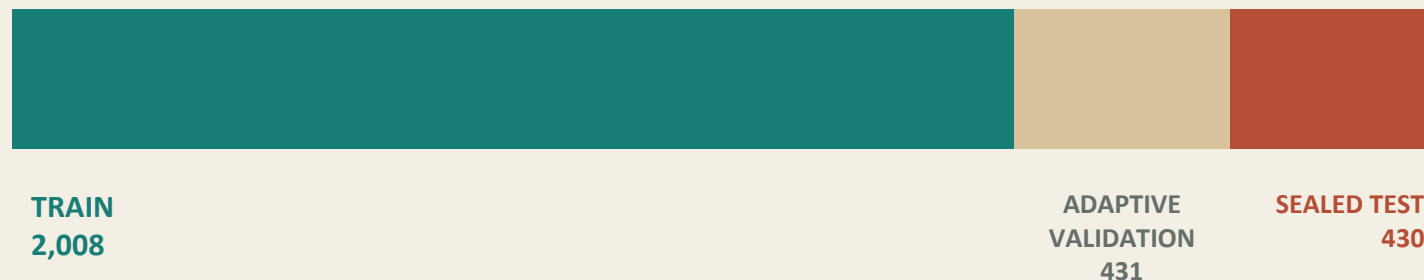
Select a module for its evidence detail. Every detail slide returns here.

Clean-v3 repairs identity leakage before model comparison

The leakage-audited manifest contains 2,869 images with no detected cross-split identity-group overlap.

2,869

leakage-audited images



IDENTITY / EXACT DUPLICATE / PERCEPTUAL DUPLICATE / SPLIT GROUP

Validation was opened adaptively. The 430-row test partition remains sealed.

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The comparison is between complete systems, not isolated encoders

MiT-B3 uses a fixed preprocessing-aware 384x384 pipeline; nnU-Net v2 uses raw RGB and a self-configuring 256x256 system.

MiT-B3 U-Net

384 × 384
LAB-luminance CLAHE
Percentile stretch + median filter
Fixed preprocessing-aware system

nnU-Net v2

256 × 256
Raw RGB
Generated plans
Self-configuring 2D system

Different resolution, decoder, loss, augmentation, and policy: this is not an isolated encoder ablation.

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nnU-Net records higher point estimates, not statistical superiority

On adaptive development-validation, nnU-Net v2 has higher robust Dice and boundary F1 point estimates; MiT-B3 retains slightly higher precision.

Robust Dice

0.8959



0.9019



Boundary F1

0.4145



0.4369



Point estimates only
Single recorded run
No paired confidence
interval

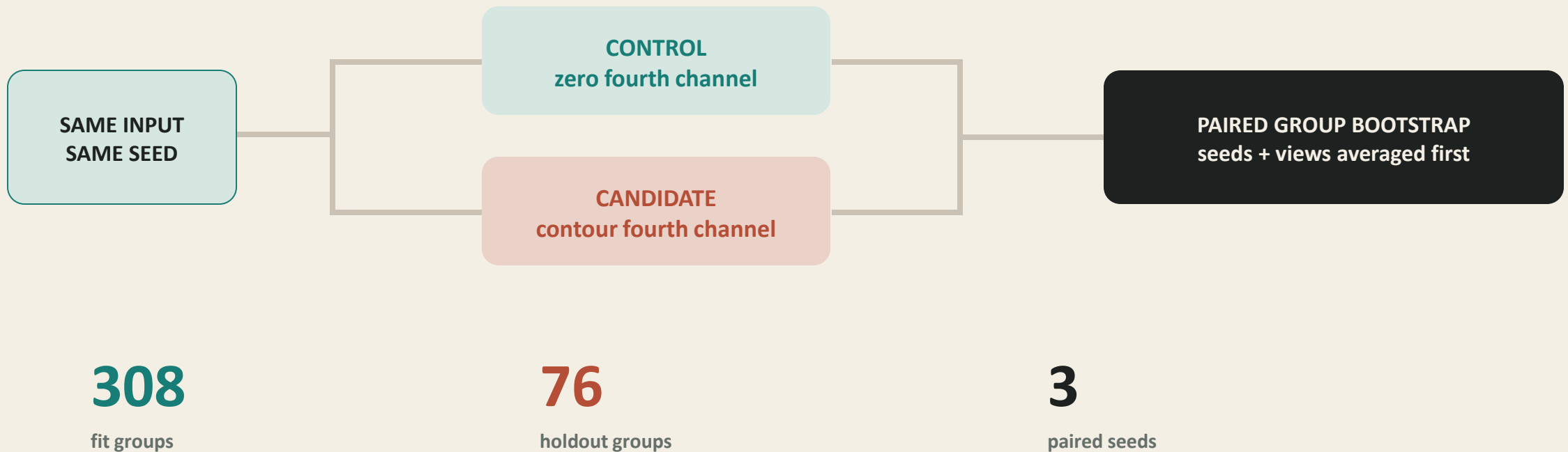
MiT-B3 retains slightly higher
precision; metric trade-offs
remain material.

MiT-B3 nnU-Net v2

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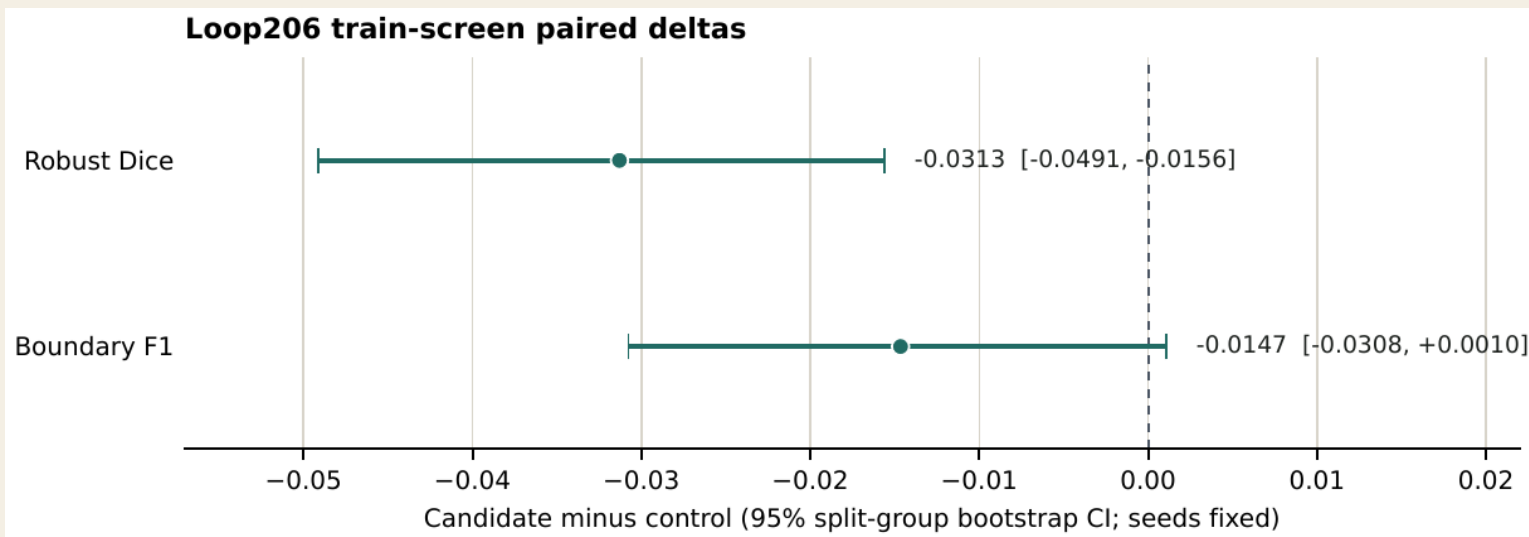
Loop206 isolates one intervention: the fourth input channel

Control and candidate share architecture, preprocessing, seeds, budget, and loss; only the zero versus contour channel changes.

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The contour channel fails the primary gate

Seed-and-view-averaged robust Dice changes by -0.0313 ; its conditional 95% interval lies entirely below zero.



REJECT

Dice Δ -0.0313
95% CI [-0.0491, -0.0156]

BF1 Δ -0.0147
95% CI [-0.0308, 0.0010]

Candidate rejected before protected evaluation.

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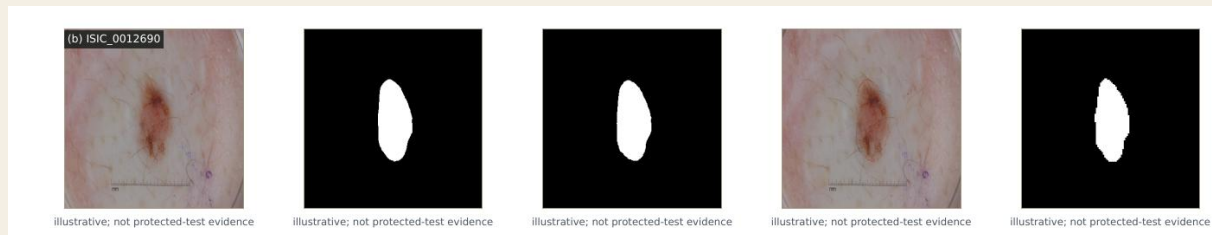
Two demo lanes answer different questions

Live public/synthetic RGB runs L206-control-s206 then reconstructed L192 nnU-Net; audited fixed samples compare the two Loop206 arms. Live output is not paper RQ1.

AUDITED FIXED L206 LANE

Fixed-cache sample: original + control mask + candidate mask

ISIC_0000050 | ISIC_0012690



illustrative fixed-cache examples; not protected-test evidence

Ground truth is authorized only in the audited fixed lane.

LIVE PUBLIC/SYNTHETIC LANE

SAME RGB, SEQUENTIAL
L206-control-s206
L192-nnUNet-v2-raw-100ep

2 public samples: illustrative masks + hash receipts
No ground truth. No accuracy. Not paper RQ1.

Non-clinical research demo; not a diagnosis.

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Can a clean score survive an invalid split?

A polished score cannot rescue a patient split whose identities cross evaluation boundaries.

PROBLEM

Leakage can make a validation split look more independent than it is.

RESPONSE

Audit identity groups and split assignments before interpreting model scores.

REMAINING LIMITATION

Audits reduce detectable leakage; they do not prove every hidden relationship is absent.

What makes a comparison fair enough to interpret?

A fair comparison needs its system differences and geometry contract stated before its ranking is discussed.

PROBLEM

Complete systems can differ in preprocessing, resolution, geometry, loss, and inference policy.

RESPONSE

Describe the comparison as system-level, then use matched arms for a causal geometry-preserving test.

REMAINING LIMITATION

A system comparison cannot identify which individual component caused a difference.

How much certainty can adaptive validation support?

Adaptive validation supports development decisions, not a certainty claim about unseen performance.

PROBLEM

Repeated development choices can make validation point estimates selection-optimistic.

RESPONSE

Label uncertainty, report the available interval, and keep the protected test sealed.

REMAINING LIMITATION

Recorded evidence has no paired confidence interval for the architecture comparison.

When should an interactive demo be trusted?

A demo earns trust by showing only the lane, inputs, and claims its evidence contract authorizes.

PROBLEM

A convincing visual demo can be mistaken for accuracy or generalization evidence.

RESPONSE

Keep illustrative live outputs separate from audited fixed comparisons and state their scope.

REMAINING LIMITATION

The demo does not establish paper RQ1 or a canonical live receipt.

What can a reproducible deployment actually prove?

Reproducibility is strongest when deployment checks distinguish verified artifacts from unavailable historical experiments.

PROBLEM

Hash-bound release artifacts do not make incomplete historical training runs clone-runnable.

RESPONSE

Verify available source bytes in the strict local audit and document deployment boundaries.

REMAINING LIMITATION

Missing code, configurations, and predictions prevent full experiment reproduction.

Artifacts are hash-bound; full experiments are not clone-runnable

Recorded claims are digest-recorded; source bytes are verified only in the strict local audit, while missing historical training artifacts keep full experiments from being clone-runnable.

HASH-BOUND

Recorded claims
Tables + figures
Compiled PDF
Model IDs
Source reports

RECONSTRUCTED RUNTIME

L192 runtime identity
Startup verification
No original equivalence
No full E2E proof

NOT CLONE-RUNNABLE

Historical training code
Loop191/192 configs
Paired predictions
RQ1-v2 pending until P1

Protected test remains sealed.

A narrower conclusion is the stronger scientific result

nnU-Net records higher development-validation point estimates, the hard contour prior fails its matched train-screen gate, and no protected-test or state-of-the-art claim is authorized.

- 01 Repair leakage before optimizing models.
- 02 Use strong baselines without hiding contract differences.
- 03 Publish negative ablations when they reject a mechanism.

NEXT DEFENSIBLE EXPERIMENT

Reproduce both systems under one geometry contract before any protected test.